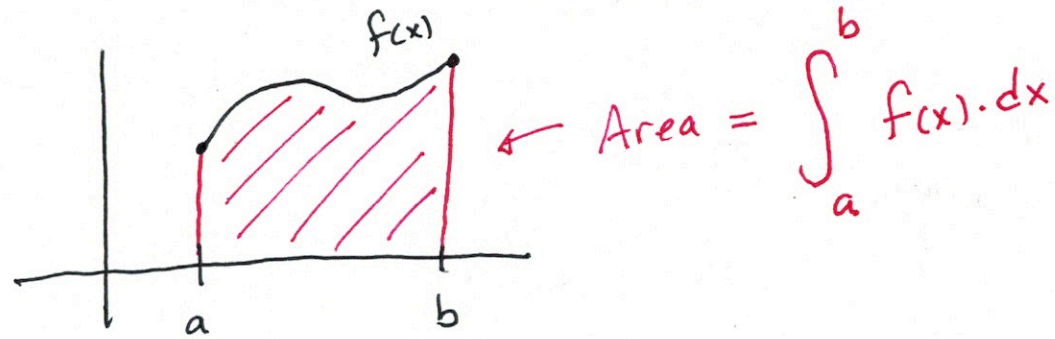


Section 5.2 The Definite Integral



FACTS

If $f(x) \geq 0$, then $\int_a^b f(x) \cdot dx \geq 0$

If $f(x) < 0$, then $\int_a^b f(x) \cdot dx < 0$

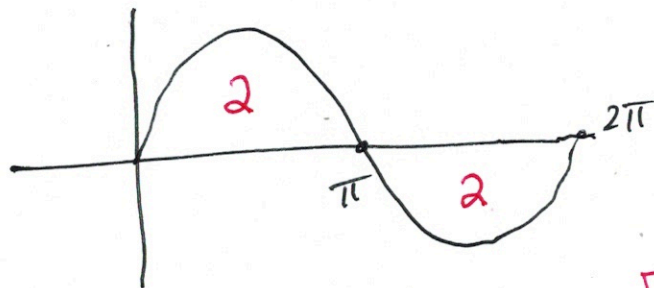
If k is a constant

$$\int_a^b k \cdot f(x) dx = k \cdot \int_a^b f(x) dx$$

$$\int_a^a f(x) dx = 0$$

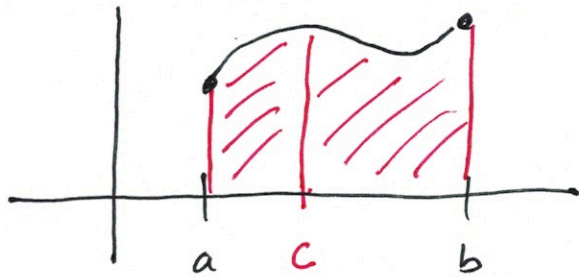
$$\text{Sum}(\text{seg}(\sin x \cdot \frac{\pi}{100}, x, \frac{\pi}{200}, \frac{199\pi}{200}, \frac{\pi}{100}))$$

$$= \boxed{2}$$



$$\text{Total area} = \boxed{4}$$

$$\int_a^b f(x) dx = - \int_b^a f(x) dx$$

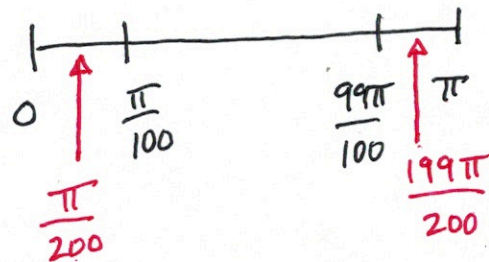


$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$

Ex ① Estimate $\int_0^{\pi} \sin x dx$ using 100 rectangles

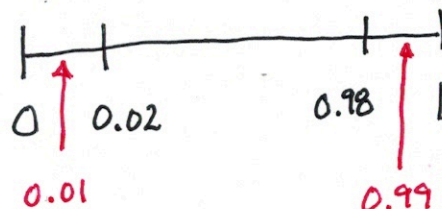
Find the area between $y = \sin x$ &
x axis on $[0, 2\pi]$.

$$\Delta x = \frac{\pi - 0}{100} = \frac{\pi}{100}$$



Ex (2) $\int_0^1 \frac{1}{1+x^2} dx$ Estimate area using
 $n = 50$ rectangles.

$$\Delta x = \frac{1-0}{50} = \frac{1}{50} = .02$$



$$\text{Sum}(\text{seq}(\frac{1}{1+x^2} \cdot (.02), x, 0.01, 0.99, 0.02))$$

$$= \boxed{0.785}$$

$$\int_0^1 \frac{1}{1+x^2} dx = \tan^{-1} x \Big|_0^1$$

$$= \tan^{-1} 1 - \tan^{-1} 0$$

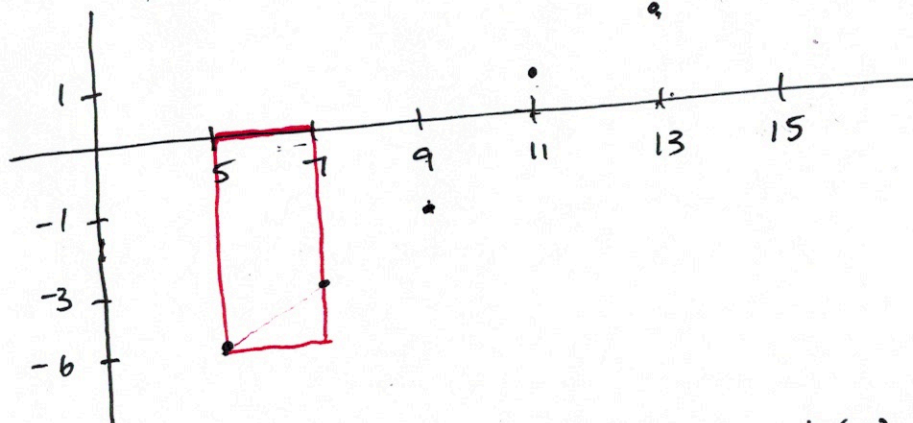
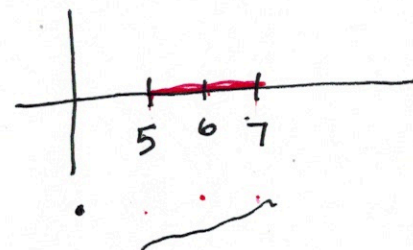
$$= \frac{\pi}{4} - 0 = \boxed{\frac{\pi}{4}}$$

Ex ③

X	5	7	9	11	13	15
f(x)	-6	-3	-1	1	5	7

Find the upper sum and lower sum

for $\int_5^{15} f(x) dx$



upper sum

$$\begin{aligned}
 & -3(2) - 1(2) + 1(2) \\
 & + 5(2) + 7(2) \\
 & = -6 - 2 + 2 + 10 + 14 = \boxed{18}
 \end{aligned}$$

$$\begin{aligned}
 \text{lower sum} &= -6(2) - 3(2) - 1(2) \\
 & + 1(2) + 5(2) \\
 & = -12 - 6 - 2 + 2 + 10 \\
 & = \boxed{-8}
 \end{aligned}$$